Gravitational potential energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate height change to near-surface gravitational energy.

SUCCESS CRITERIA - I can...

1. Choose a height reference.
2. Use ΔU=mgΔh with the stated g.
3. Distinguish potential energy from force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “potential energy” and “reference level” to explain your answer. Sentence starters are optional.

potential energy: energy associated with the configuration of interacting objects

reference level: the chosen position where potential energy is set to zero

weight: the gravitational force on an object

01 Warm-up: bring back an earlier idea

A cart’s kinetic energy changes from **15 J** to **3 J**. **Find net work** and state whether its speed rises or falls.

Word help: kinetic energy = energy associated with motion

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

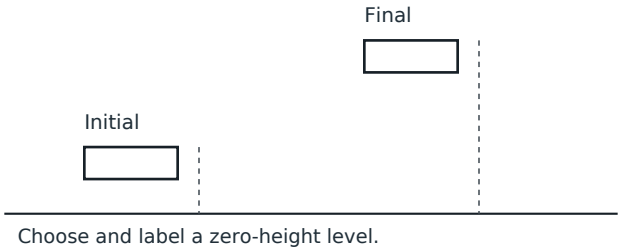
02 Worked example: explain the prediction

Reference: $\Delta U = mg\,\Delta h = mg(h_f - h_i)$.
Approximately constant g = **10 N/kg**. **Choose a zero height**; the energy change does not depend on that choice.

Worked reasoning: Raising a **2 kg** object by **3 m** with g=**10 N/kg** increases gravitational potential energy by mgh=**60 J**. The chosen zero does not affect this change.

Your turn: Draw initial and final height levels and **label the 3 m** difference.

Sentence starter: Both ... . They differ in ... because ... .



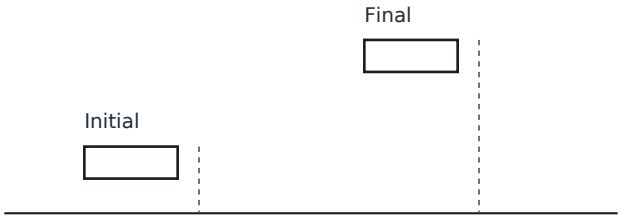
Choose and label a zero-height level.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two identical books sit at heights **1** and **3 m** above the floor. Which has more gravitational potential energy relative to the floor?

Sentence starter: I notice ... . This supports ... because ... .



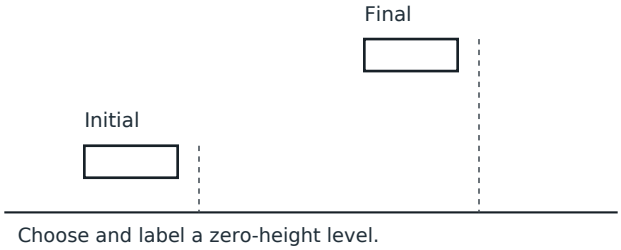
Choose and label a zero-height level.

Marzano check - circle: ① ② ③

04 Apply the physics

Each book in question **3** has mass **2 kg**. Use g = **10 N/kg** and U = **0** at the floor. **Find U** at **1 m** and at **3 m**. Then find ΔU when one book is lifted from **1 m** to **3 m**. **Mark the vertical change**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Choose and label a zero-height level.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does a negative value of U relative to a chosen zero imply a negative mass or impossible height?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Equal-mass carts have velocities **+6** and **-4 m/s** before an ideal **1** D elastic collision. **Give both final velocities**. Use before and after snapshots to organize the momentum account.

Word help: elastic = a collision that conserves total kinetic energy

Sentence starter: I notice … . This supports … because … .

Before	After

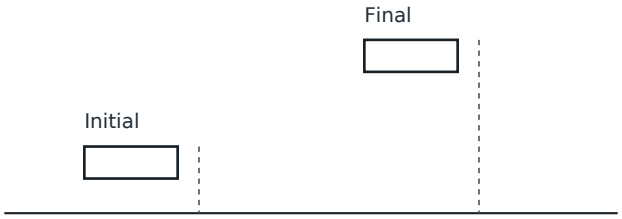
Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

Lift a **3 kg** object by **2 m**.
1. Find ΔU with g=**10**.
2. **Label the relevant heights or** distances on the diagram.

Sentence starter: I found … [unit]. This value means … in this situation.



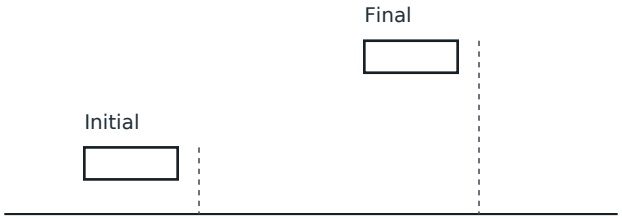
Choose and label a zero-height level.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Construct two mass-and-height-change pairs with the same **60 J** increase using g=**10**. **Label the relevant heights or** distances on the diagram.

Sentence starter: In my work, I used … to show … . I labeled it … .



Choose and label a zero-height level.

Marzano check - circle: 1 2 3